

BAAZ JKBOSE



2026 CLASS-11TH

Nueral Control and
Coordination ✓

LECTURE-2

Zoology

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Topics To Be Covered

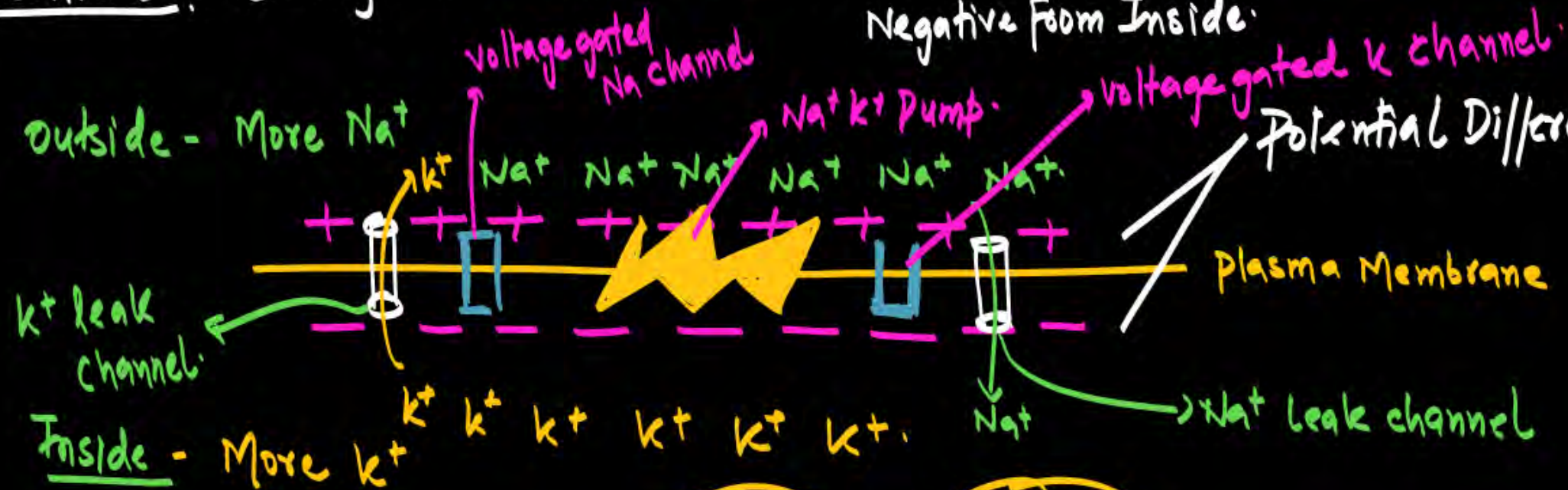
- 1 Generation and Conduction Of action potential

Generation & Conduction of Nerve Impulse / Signal / Action potential.

→ A neuron Membrane during Resting is in polarised State.



① polarisation - During this State, Membrane is Positive from outside
Negative from Inside.



Minus. (inside)

Potential Difference: Resting = -70mV.
Membrane potential

- More Negatively charged protein, PO_4^{2-}

PO_4^{2-}

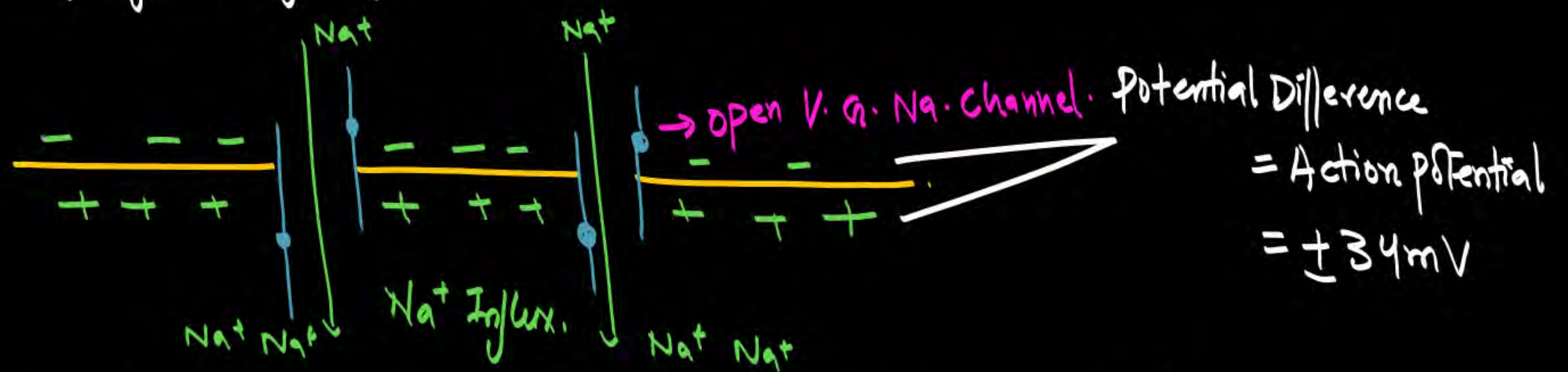
protein (-)

→ The Membrane has three types of channel (Protein Tunnels).

- i) Na^+/K^+ Leak channels. (Through them Na^+ & K^+ leaks).
- ii) Na^+/K^+ Pump. (Uses Energy to take 3 Na^+ outside and 2 K^+ Inside)
- iii) Na^+ and K^+ Voltage Gated channels. (Sensitive to voltage and have gates)

b) Depolarisation. - When a strong Stimulus Hits Membrane

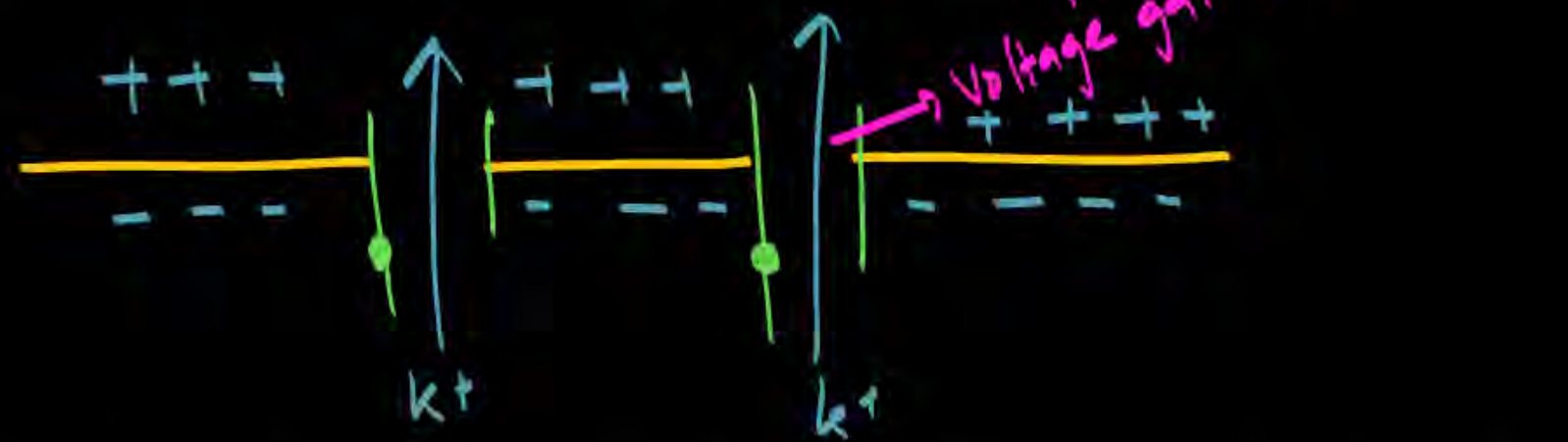
- opening of Voltage gated Na^+ channels



- Due to Na^+ Influx, the Membrane from Inner Side Becomes positive.

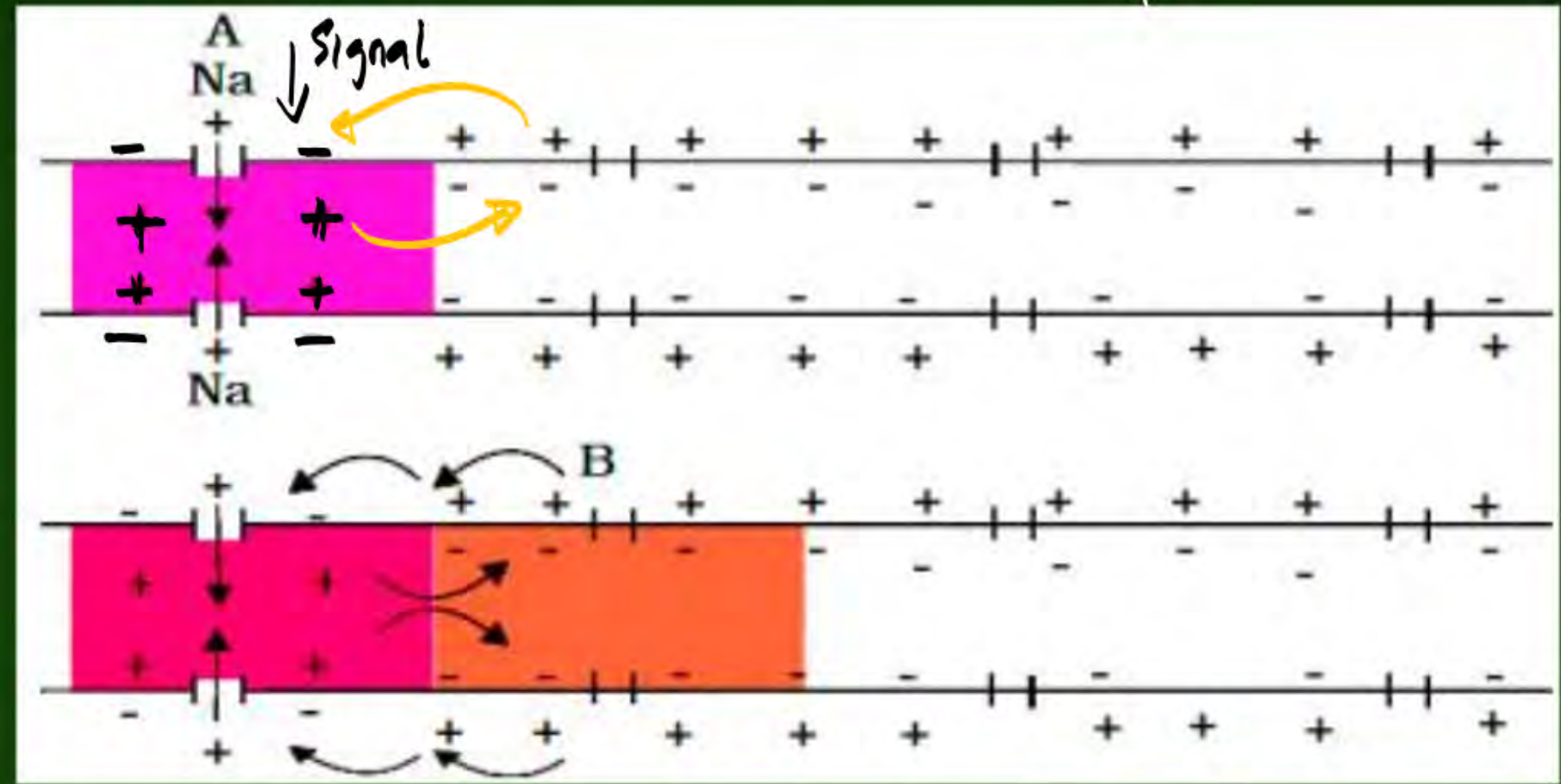
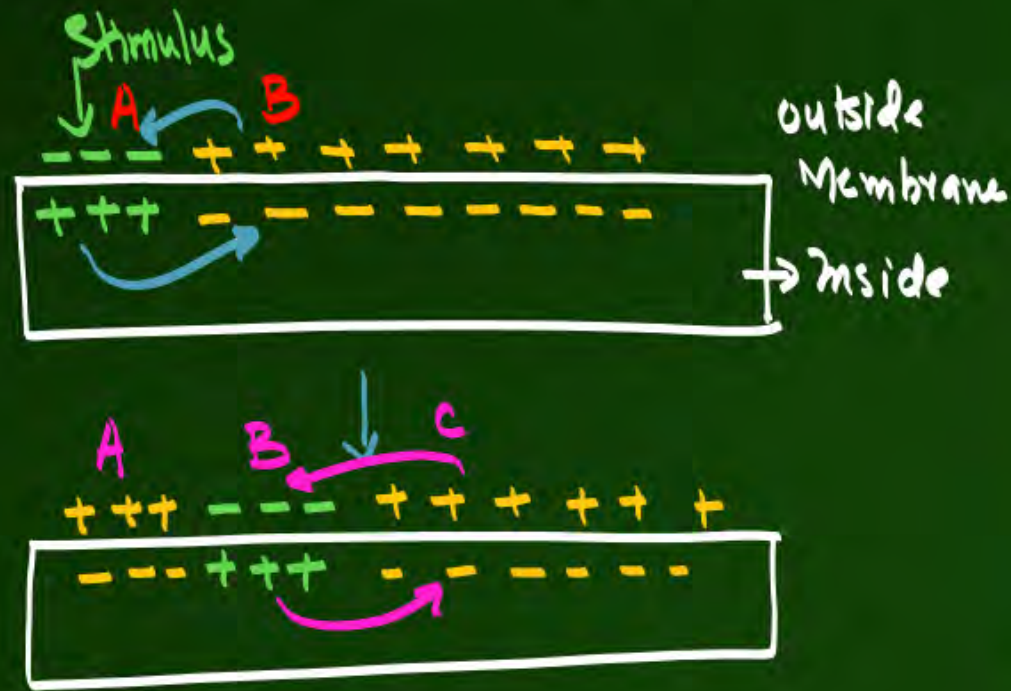
② Repolarisation

- Here the Resting potential is Restored
- It is By opening of voltage gated K^+ channels.



→ It cause outflux of K^+ , Restoring positive charge outside.

(Travel).
→ Conduction of Action potential / Impulse



- The depolarisation at point 'A' is conducted to point 'B' and then to 'C' and so on.
- The flow of current/charge is A to B From Inside and B to A From outside

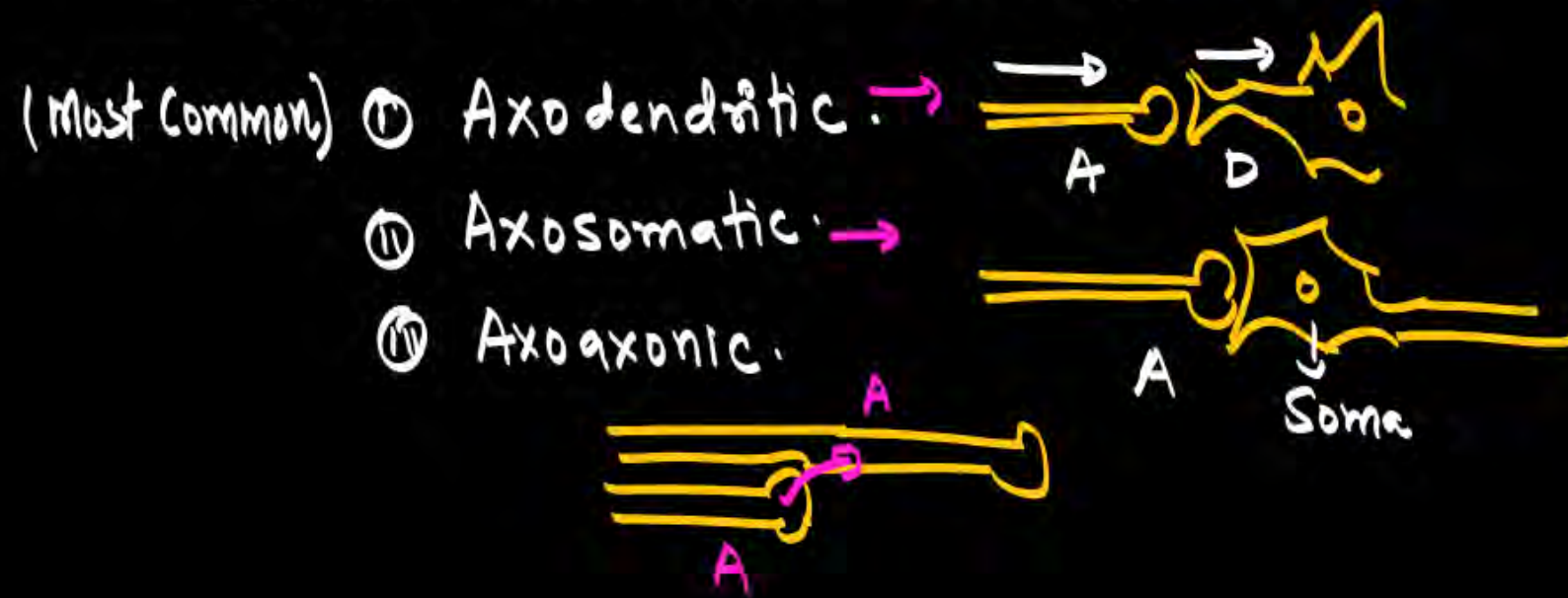
++++
++
+2
-1
-1

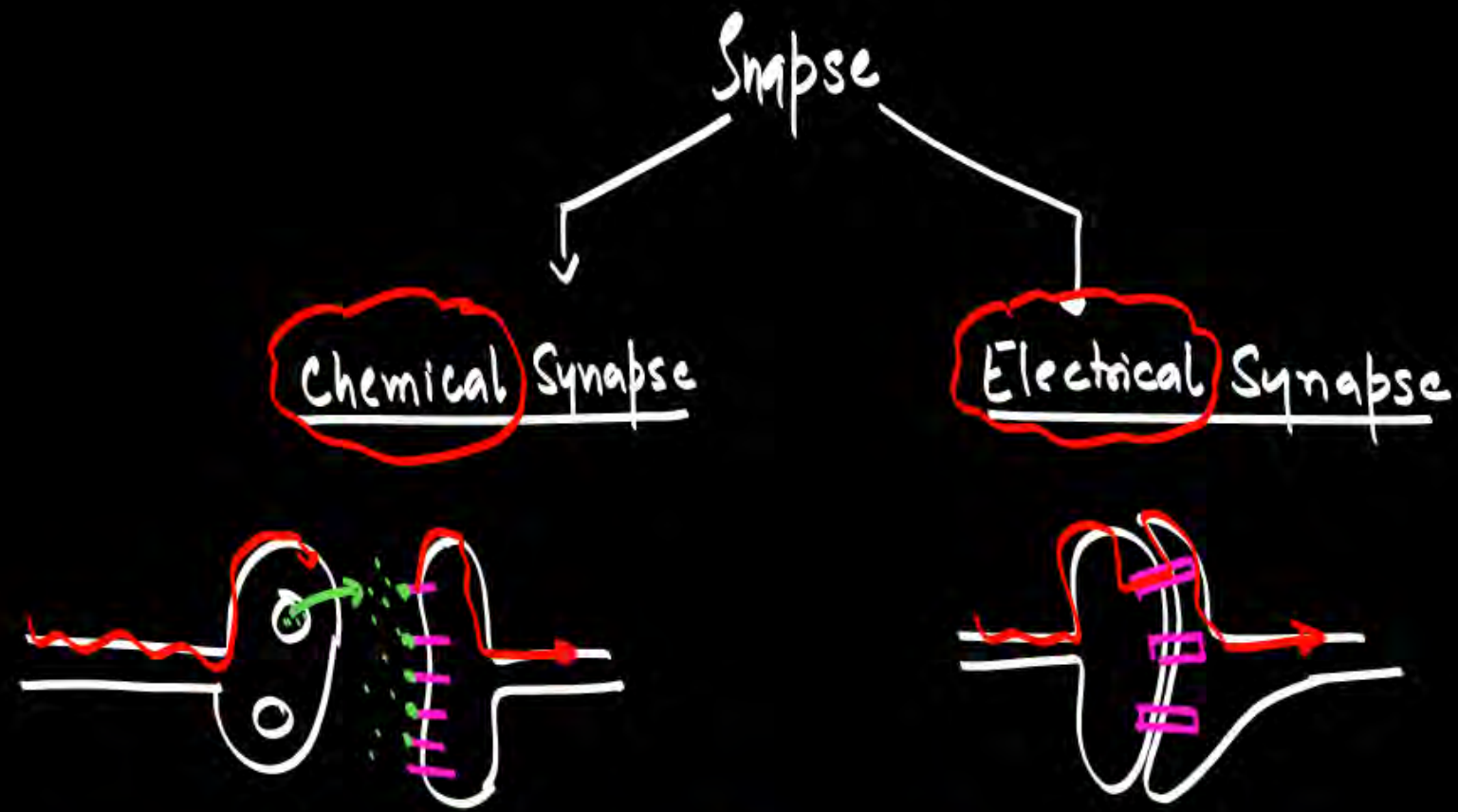
Synapse

↳ It is a junction at which impulse is transferred from one cell to another.
(signal)

- Can be btw two neurons, Neuron & Muscle cell, Muscle & Muscle cell.

- Between Neurons, it can be;





- Gap Btw cell is More (Synaptic cleft)
- Here chemicals (Neurotransmitters) are Required
- Slow (Because of synaptic Delay).
- Mostly in our Nervous System.

- Synaptic cleft is less, cell are joined by Gap junctions.
- Signal transfers through gap junctions.
- Quick.
- very less in Nervous System.

Types of Neurotransmitters

Excitatory

- Excite the Another cell. (ON)

(e.g) - Acetylcholine

- Adrenaline

- Nor Adrenaline

Inhibitory

- Inhibit the Signal in Another cell. (OFF).

- Dopamine (Deficiency - Parkinson's Syndrome)

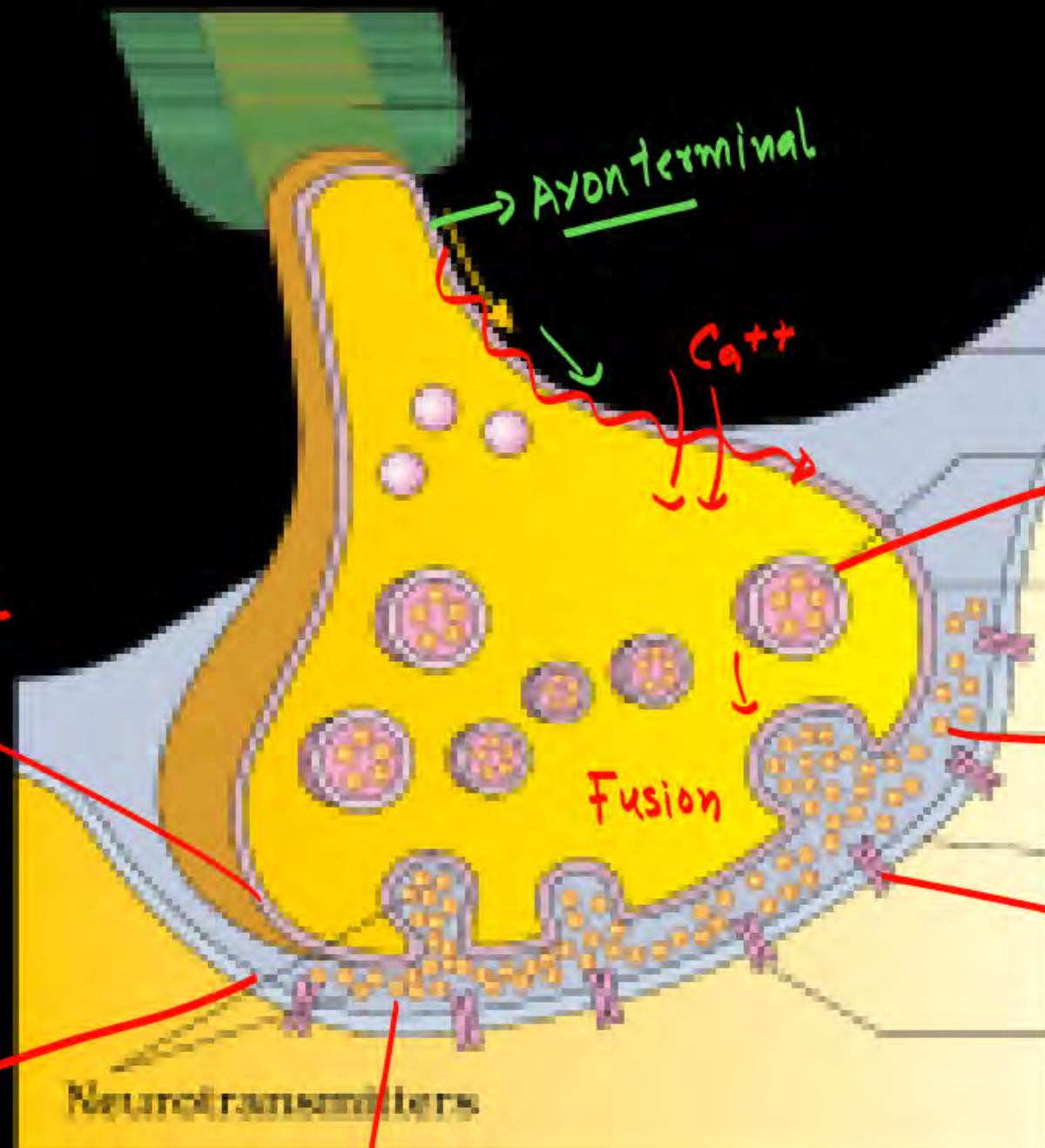
- Glycine

- GABA - Gamma Amino butyric Acid.



Pre Synaptic Membrane

Post Synaptic Membrane



Axon terminal

Ca^{++}

Fusion

Neurotransmitters

Synaptic Vesicle

Synapse Diagram

Release of Neurotransmitter

Receptors

Synaptic cleft (gap)

Thank
You